GPU-Accelerated Density Functional Theory Solver for Binary SALR Colloids (SALR3YUCUDA)

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***Abstract*—We present a computational framework for solving mean-field density functional theory (DFT) equations describing spatial density distributions of binary colloidal mixtures with competing short-range attraction and long-range repulsion (SALR) interactions. The solver implements the Picard iterative method for a two-component system with different walls configuration, using the triple Yukawa potential model. Our CPU implementation employs OpenMP parallelization, achieving significant speedup on multi-core processors. A CUDA implementation for GPUs was developed, delivering a multi-fold speedup over the CPU baseline; on the default benchmark, the GPU run was $5.87\times$ faster than a 20-thread OpenMP execution. We demonstrate the solver capabilities by computing equilibrium density profiles under periodic, two-wall, and four-wall boundary conditions. A Qt-based graphical user interface (GUI) was also built to facilitate parameter configuration, task execution, and result visualization.**

I. Introduction

Colloidal systems provide important models for studying fundamental phenomena in condensed matter physics, including phase transitions, self-assembly, and complex fluid behavior [1]. Among the most intriguing colloidal systems are those exhibiting competing interactions, where particles experience short-range attraction and long-range repulsion (SALR). These interactions, sometimes called "mermaid" potentials due to their attractive head and repulsive tail [1], lead to rich phase behavior including microphase separation, cluster formation, and periodic stripe patterns [2]. Figure 1 illustrates a typical SALR potential profile, with the short-range attractive well ("head") followed by a long-range repulsive barrier ("tail").

Despite significant theoretical predictions about SALR systems, experimental realization of the predicted mesophases in three-dimensional real space remains challenging. As Royall notes, "no ordered phase in a system with competing interactions has ever been observed in 3d real space" [1]. This gap between theory and experiment motivates the development of efficient computational tools that allow researchers to explore parameter space rapidly and identify conditions under which specific morphologies may emerge.

The theoretical description of spatial density distributions in SALR systems is commonly performed using density functional theory (DFT) within the mean-field approximation [3]. This approach leads to a system of nonlinear integral equations that must be solved iteratively. For a two-dimensional grid of $N_x \times N_y$ nodes, the computational cost per iteration scales as $O(N^2)$ due to the convolution required to compute interaction fields, making the problem computationally intensive.

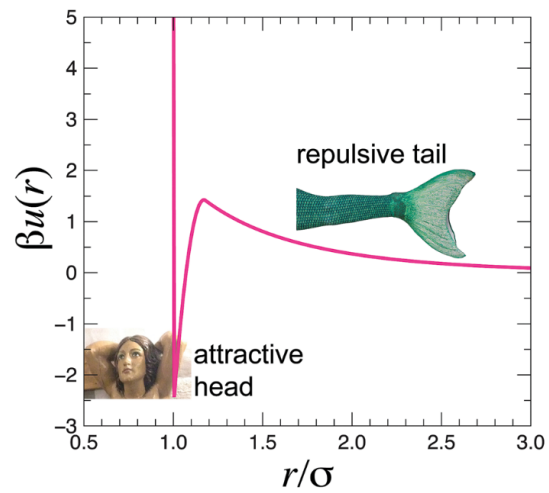

Fig. 1. A "mermaid" (SALR) pair potential as a function of reduced separation r/σ. The attractive head promotes local condensation while the repulsive tail prevents bulk phase separation, driving the system toward microphase-separated morphologies such as clusters and stripes. Reproduced from [1].

In this work, we present SALR3YUCUDA - a solver for mean-field DFT equations describing binary SALR mixtures. The project name reflects its key components: SALR (Short-range Attraction Long-range Repulsion), 3Y (triple Yukawa potential), and CUDA (GPU acceleration technology). We implemented an optimized pure C version with OpenMP parallelization for scalable execution on multi-core processors, accompanied by a highly parallelized CUDA framework for GPU-accelerated massive parameter computations. The solvers natively integrate with an SQLite-HDF5 database for session tracking and large spatial tensor storage, and feature multiple boundary conditions including periodic boundaries (PBC), two parallel walls (W2), and four-wall confinement (W4). Furthermore, a Qt-based graphical interface enables rich visual exploration without cumbersome command-line workflows.

A. Motivation

Binary colloidal mixtures with SALR interactions are of fundamental interest across soft matter, biophysics, and materials science. The interplay of competing attraction and repulsion gives rise to self-assembled mesophases—clusters, stripes, and lamellar structures—that serve as model systems for understanding protein aggregation, lipid membranes, and block-copolymer microphase separation [1], [2]. Mapping the conditions under which each morphology appears requires systematic exploration of a high-dimensional parameter space (composition, temperature, interaction strengths and ranges), which is experimentally costly and time-consuming.

Particle-resolved computer simulations such as molecular dynamics (MD) and Monte Carlo (MC) provide the most detailed description of these systems, tracking every particle trajectory and configuration. However, their computational cost scales with the number of particles N_p: a single equilibration run for a system large enough to contain several stripe periods may require millions of MC sweeps or nanoseconds of MD trajectory, making large-scale parameter surveys impractical. Furthermore, finite-size effects are pronounced when the system box is not much larger than the characteristic pattern wavelength.

Density functional theory offers a complementary approach that avoids these limitations. Rather than evolving individual particle positions, DFT works directly with continuous density fields $\rho_i(\mathbf{r})$, reducing an N_p-body statistical mechanics problem to a deterministic nonlinear integral equation solved on an $N_x \times N_y$ grid. Key advantages include:

- **Direct equilibrium access:** The iterative solver finds the equilibrium density profile directly without the need for thermalization or ensemble averaging inherent to stochastic simulations.
- **Clean boundary conditions:** Walled geometries are imposed analytically as Dirichlet constraints on ρ_i, avoiding surface-energy artifacts from finite particle layers.
- **Parameter scalability:** Changing temperature, density, or interaction coefficients requires only reinitializing the solver, enabling rapid sweeps across the phase diagram.

These properties make mean-field DFT the method of choice for initial reconnaissance of SALR phase behavior, providing guidance for more expensive particle-level studies and for comparison with theory. The present work implements and accelerates this approach, with GPU parallelization targeting the dominant $O(N^2)$ convolution kernel to rapidly expand the accessible parameter space.

II. Theoretical Background

A. SALR Interactions

Short-range attraction and long-range repulsion (SALR) interactions arise naturally in colloidal systems where particles experience depletion attraction at short distances (due to polymer or small particle exclusion) and electrostatic repulsion at longer distances [1]. The competition between these interactions leads to complex energy landscapes and the possibility of microphase separation, where the system forms periodic structures such as clusters, stripes, or lamellae rather than undergoing bulk phase separation.

B. Triple Yukawa Potential

We model the pair interaction between particles of species i and j using a triple Yukawa form:

$$U_{ij}(r) = \sum_{m=1}^{3} A_{ij}^{(m)} \frac{\exp(-\alpha_{ij}^{(m)} r)}{r}, \quad 0 < r \leq r_c \tag{1}$$

where $A_{ij}^{(m)}$ are amplitude coefficients, $\alpha_{ij}^{(m)}$ are inverse screening lengths, and r_c is a cutoff radius beyond which $U_{ij}(r) = 0$. The three Yukawa terms allow flexible representation of the SALR shape: positive amplitudes at short inverse lengths create repulsion, while negative amplitudes create attraction.

For a binary mixture (species 1 and 2), we require three independent potentials: $U_{11}(r)$ for species 1–1 interactions, $U_{22}(r)$ for species 2–2 interactions, and $U_{12}(r) = U_{21}(r)$ for cross-species interactions.

C. Mean-Field Density Functional Theory

Within the mean-field approximation, the grand thermodynamic potential functional for the binary mixture leads to Euler-Lagrange equations that determine the equilibrium density profiles $\rho_1(\mathbf{r})$ and $\rho_2(\mathbf{r})$:

$$\rho_1(\mathbf{r}) = \rho_{1,b} \exp[-\beta\,(\Phi_{11}(\mathbf{r}) + \Phi_{12}(\mathbf{r}) - \Phi_{11,b} - \Phi_{12,b})] \tag{2}$$

$$\rho_2(\mathbf{r}) = \rho_{2,b} \exp[-\beta\,(\Phi_{21}(\mathbf{r}) + \Phi_{22}(\mathbf{r}) - \Phi_{21,b} - \Phi_{22,b})] \tag{3}$$

where $\beta = 1/(k_B T)$ is the inverse temperature (with k_B the Boltzmann constant), $\rho_{i,b}$ is the bulk (average) density of species i, and $\Phi_{ij}(\mathbf{r})$ are interaction field contributions defined as convolutions:

$$\Phi_{ij}(\mathbf{r}) = \int d\mathbf{r}'\, \rho_j(\mathbf{r}')\, U_{ij}(|\mathbf{r} - \mathbf{r}'|) \tag{4}$$

The bulk field values $\Phi_{ij,b}$ represent the interaction fields in a uniform system with densities $\rho_{1,b}$ and $\rho_{2,b}$, ensuring that the trivial homogeneous solution satisfies the equations.

D. Picard Iteration

Equations (2)–(3) are solved iteratively using the Picard method with mixing:

$$\rho_i^{(t+1)}(\mathbf{r}) = \xi_i K_i[\rho^{(t)}](\mathbf{r}) + (1 - \xi_i)\rho_i^{(t)}(\mathbf{r}) \tag{5}$$

where $K_i[\rho]$ denotes the right-hand side of the Euler-Lagrange equation and $\xi_i \in (0, 1]$ is a mixing parameter that controls convergence stability. Iterations continue until the root-mean-square change falls below a tolerance ε:

$$\left\| \rho_i^{(t+1)} - \rho_i^{(t)} \right\| = \sqrt{\frac{1}{N_x N_y} \sum_k \left[{\rho_i^{(t+1)}}_k - {\rho_i^{(t)}}_k \right]^2} < \varepsilon \tag{6}$$

E. Boundary Conditions

The solver supports three boundary modes:

- **PBC (Periodic Boundary Conditions):** The domain wraps around in both x and y directions, simulating a bulk-like infinite system.
- **W2 (Two Walls):** Hard walls at $x = 0$ and $x = L_x$ with periodic boundaries in y. The density is forced to zero at wall positions: $\rho_i(x = 0) = \rho_i(x = L_x) = 0$.
- **W4 (Four Walls):** Hard walls on all boundaries, confining particles to a rectangular box.

III. Implementation

A. Discretization

The two-dimensional domain $[0, L_x] \times [0, L_y]$ is discretized on a uniform grid with $N_x \times N_y$ nodes and spacing Δx, Δy. The continuous convolution integral (4) becomes a discrete sum:

$$\Phi_{ij}(x_i, y_j) = \Delta x \Delta y \sum_{m,n} \rho_j(x'_m, y'_n)\, U_{ij}(|\mathbf{r}_{ij} - \mathbf{r}'_{mn}|) \quad (7)$$

For efficiency, the potential $U_{ij}(r)$ is precomputed and stored in lookup tables indexed by displacement (d_{ix}, d_{iy}) between grid cells.

B. Algorithm Overview

The solver proceeds through the following steps:

1) **Initialization:** Load parameters; build potential tables U_{11}, U_{12}, U_{22}; compute bulk fields $\Phi_{ij,b}$; set initial density profiles with small perturbations.
2) **Iteration loop:**
   a) Compute interaction fields Φ_{11}, Φ_{12}, Φ_{21}, Φ_{22} via discrete convolution.
   b) Evaluate K_1 and K_2 using equations (2)–(3).
   c) Apply mass conservation renormalization.
   d) Perform Picard mixing (5).
   e) Apply boundary conditions (zero density at walls).
   f) Apply anti-aliasing smoothing to suppress numerical artifacts.
   g) Check convergence criterion (6).
3) **Output:** Save converged density profiles.

C. CPU Parallelization

The most computationally expensive operation is the $O(N^2)$ convolution for computing Φ_{ij}. We employ OpenMP to parallelize:

- Potential table construction (loop over grid offsets)
- Convolution computation (nested loops over source and field cells)
- K-operator evaluation and Picard mixing

On a multi-core processor (Intel Core i7-13700H, 20 threads), this OpenMP parallelization achieves a multi-fold speedup compared to single-threaded execution on the largest grid sizes tested.

D. CUDA Implementation

GPU acceleration explicitly targets the critical convolution operation by offloading both potential generation and density mixing to the CUDA kernel pipeline. We employ coalesced global memory access and constant memory for potential parameters (such as the Yukawa amplitudes and inverse screening lengths) to maximize global bandwidth throughput. Custom 2D thread blocks iteratively reduce memory fetches during the spatial convolution loops.

E. GUI and Session Database

The project integrates a Qt 6-based graphical interface for end-to-end simulation lifecycle management. Parameters such as temperature, grid shape, and boundary constraints are configured through the GUI, while the backend stores sessions and density snapshots in SQLite and HDF5 for later inspection and reuse.

IV. Results

A. Simulation Parameters

We demonstrate the solver using the parameters listed in Table I. These values match the current default configuration used by the code and the generated benchmark figures.

TABLE I
Simulation Parameters

Parameter	Value
System size $L_x \times L_y$	16.0×16.0
Grid resolution $N_x \times N_y$	160×160
Grid spacing $\Delta x = \Delta y$	0.1
Temperature T	8.0
Bulk density $\rho_{1,b}$	0.4
Bulk density $\rho_{2,b}$	0.2
Cutoff radius r_c	8.0
Mixing coefficients $\xi_1 = \xi_2$	0.02
Maximum iterations	50000
Tolerance ε	10^{-8}

The triple Yukawa coefficients for species 1–1 interaction are $A_{11}^{(1)} = 150.66$, $A_{11}^{(2)} = -122.61$, $A_{11}^{(3)} = 27.12$ with corresponding $\alpha_{11}^{(1)} = 1.92$, $\alpha_{11}^{(2)} = 1.26$, $\alpha_{11}^{(3)} = 0.76$. These parameters produce a potential with attractive head and repulsive tail characteristic of SALR systems.

B. Periodic Boundary Conditions (PBC)

Figure 2 shows the converged density distribution under periodic boundary conditions. The system exhibits stripe formation, with alternating regions of high species-1 density (purple) and high species-2 density (green). This demixing pattern reflects the competition between short-range attraction (favoring condensation) and long-range repulsion (preventing bulk separation).

C. Two-Wall Confinement (W2)

Figure 3 presents results for confinement between two parallel walls at $x = 0$ and $x = L_x$. The hard-wall boundary condition ($\rho = 0$ at walls) fundamentally alters the density distribution compared to PBC:

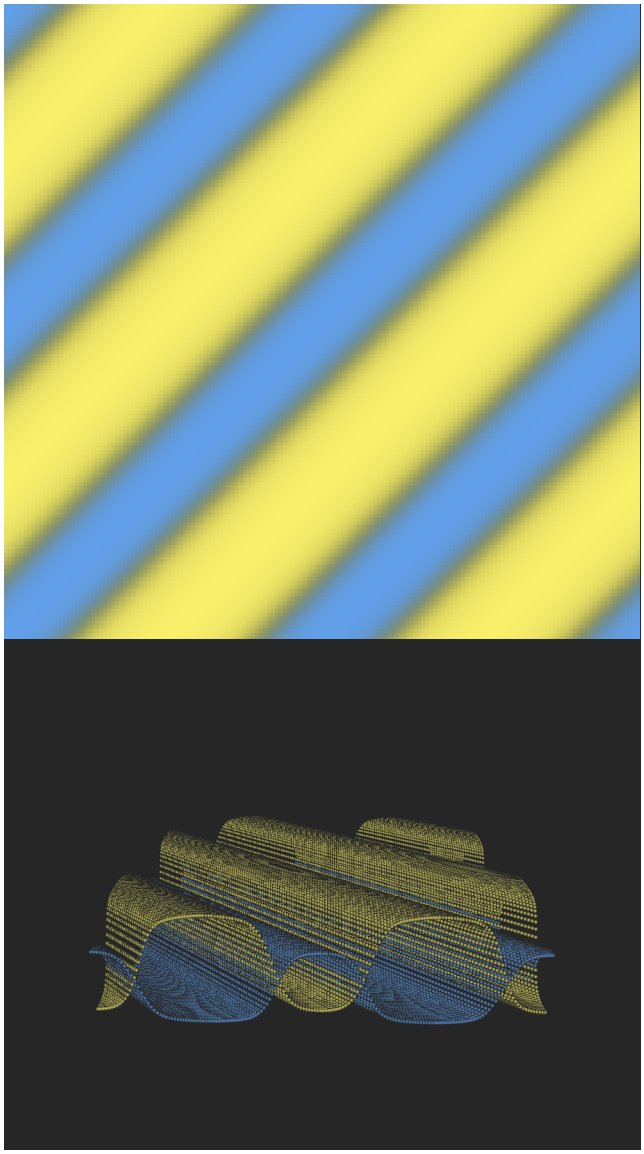

Fig. 2. Density distribution under periodic boundary conditions (PBC). Top: 2D heatmap of the final density field. Bottom: 3D scatter view of the same run. Stripe-like modulations indicate microphase separation on the periodic domain.

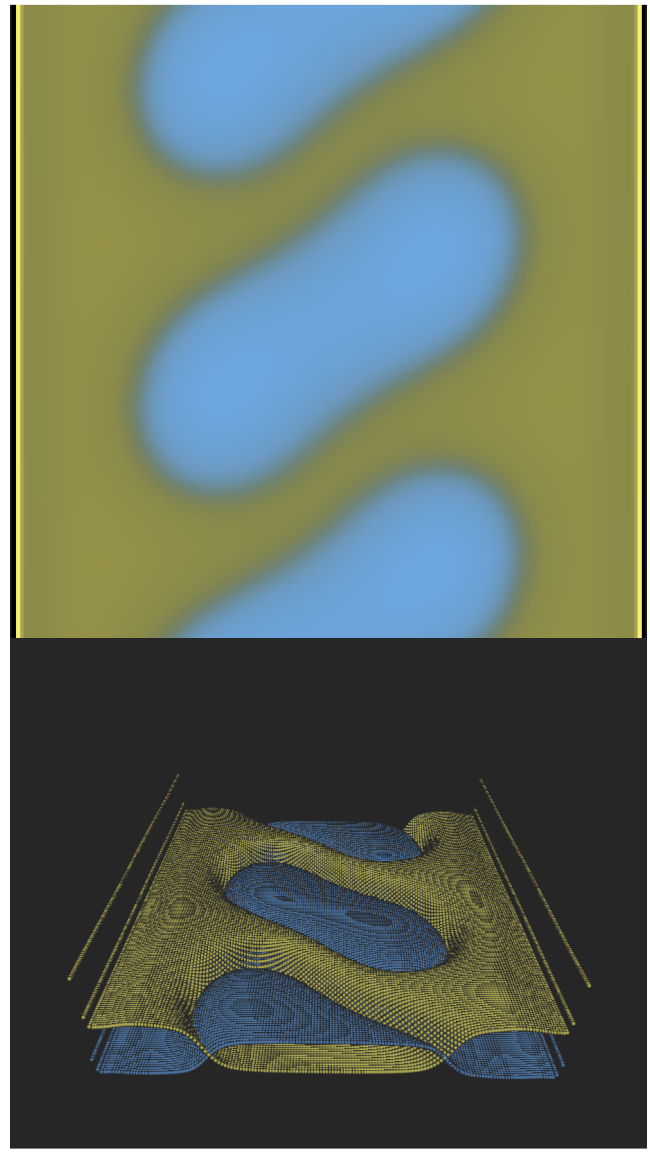

Fig. 3. Density distribution with two-wall confinement (W2). Top: 2D heatmap showing depletion near the walls and wall-parallel modulation. Bottom: 3D scatter view of the same confined state.

- Density depletes near the walls, creating boundary layers where both species have reduced concentration.
- Stripes orient parallel to the walls (perpendicular to the confinement direction), as this orientation minimizes the free energy penalty from wall interactions.
- The periodic direction (y) allows stripe continuity, producing a layered structure.

D. Four-Wall Confinement (W4)

Figure 4 shows the fully confined case with hard walls on all four boundaries. The box geometry produces:

- A depleted boundary layer surrounding the entire perimeter.
- Complex internal structure with density concentrated in the central region.
- Corner effects where the two perpendicular walls create regions of severe depletion.
- Species segregation within the confined volume, with the higher-density species (species 1) forming a more compact central distribution.

E. Performance

A scalability analysis details the runtime throughput for the default parameter set. On the 160×160 grid, the 20-thread OpenMP run converged in 7647.405s, while the CUDA version converged in 1302.876s with the same final error and iteration count. This corresponds to a $5.87\times$ wall-time speedup for CUDA on the benchmark used in the generated figures.

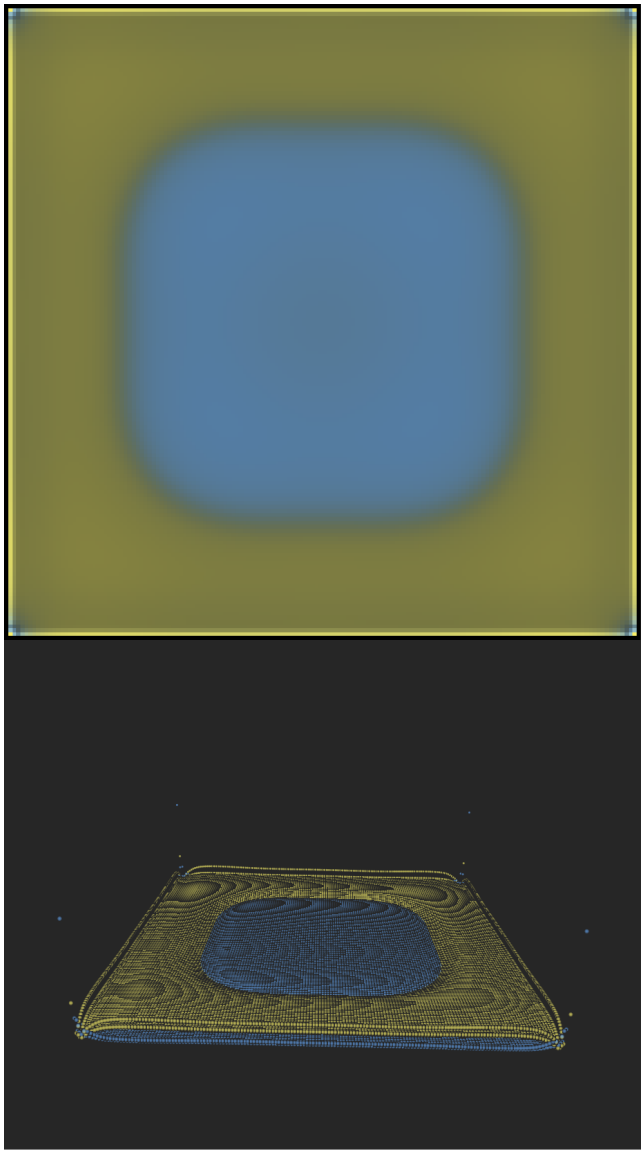

Fig. 4. Density distribution with four-wall confinement (W4). Top: 2D heatmap showing boundary depletion on all sides and central concentration. Bottom: 3D scatter view of the same confined state.

V. Discussion

The computed density distributions demonstrate that our solver correctly captures the essential physics of SALR systems: microphase separation driven by competing interactions, and the influence of confinement on pattern morphology.

The observation of vertical stripes under PBC and wall-parallel stripes under W2 confinement aligns with theoretical expectations. In the bulk-like PBC case, symmetry is broken by the initial conditions, selecting a particular stripe orientation. When walls are introduced, the system minimizes its free energy by aligning stripes parallel to the confining surfaces, reducing the density gradient penalty at the boundaries.

These results support the utility of computational tools for exploring parameter space in SALR systems. As noted by Royall [1], experimental realization of predicted mesophases faces challenges including weak ion dissociation in non-aqueous solvents and non-centrosymmetric interactions. Simulations in idealized conditions allow researchers to establish baseline expectations before confronting experimental complexities.

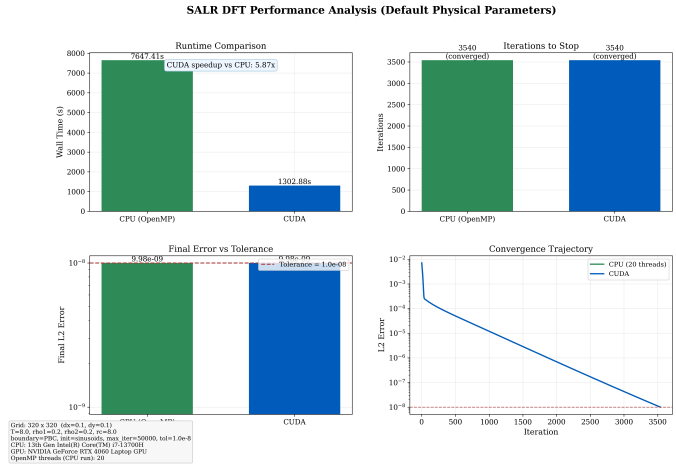

Fig. 5. Convergence-focused CUDA vs. CPU(OpenMP) benchmark for the default physical parameters. Both runs reach the same tolerance, but the GPU version completes substantially faster.

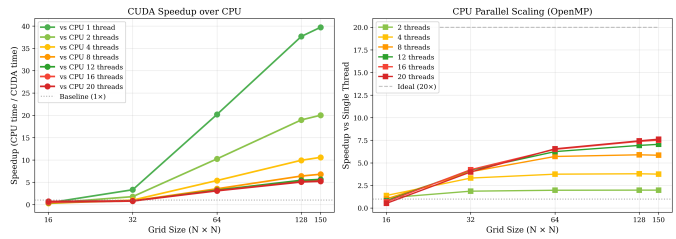

Fig. 6. Scaling analysis comparing CUDA speedup against CPU baselines across grid sizes and OpenMP thread counts.

Specifically, the deployment on CUDA reduces the runtime of the default benchmark from more than two hours on the 20-thread CPU run to a little over twenty minutes on the GPU. The scaling plot also shows that the GPU remains competitive as the grid size grows, while the CPU version depends more strongly on thread count and memory bandwidth.

VI. Conclusion

We have presented SALR3YUCUDA, a computational solver for mean-field DFT equations describing binary SALR colloidal mixtures. The CPU implementation with OpenMP parallelization and the CUDA accelerator both compute equilibrium density distributions under periodic and confined boundary conditions, capturing the stripe-like morphologies expected for competing interactions.

The current benchmark set uses the default 160×160 configuration from the code base. On that case, CUDA reached convergence $5.87\times$ faster than the 20-thread CPU baseline, and the report now includes the matching convergence and scaling plots. Furthermore, the integrated Qt GUI serves as a practical pipeline for iterative testing and on-the-fly visualization of distributions.

Future work will expand horizontally along several vectors including:

- Extension to true three-dimensional domains.
- Multi-GPU scaling via deep domain decomposition.
- Automated phase-diagram extraction algorithms mapping extensive structural regions seamlessly across the interactive GUI.

ACKNOWLEDGMENT

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